

Date	3 July 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current AI-Driven Emotion Detection & Personalized Learning Support Platform can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Limited support for concurrent users	System performance may degrade with many learners accessing simultaneously	High
2	CSV-based storage does not scale well as interaction history grows	Slower lookups and no clean multi-user data isolation as usage increases	High
3	No persistent user accounts or long-term progress tracking	Learners cannot access their emotional history across sessions or devices	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Runs as a single local Streamlit session	Deploy on cloud infrastructure (AWS, Google Cloud, Azure) with load balancing and auto-scaling for multiple concurrent learners
2	Data Storage	Interaction history and feedback stored in local CSV files	Migrate to a scalable database such as PostgreSQL or MongoDB with cloud storage support
3	Performance	Emotion classification and Gemini guidance generated in real-time for each request	Use caching, batching, and optimized model inference pipelines to reduce latency under load
4	Security	Basic local storage with no authentication	Implement secure user authentication, encrypted storage, and secure API key management for Gemini calls

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Integration with LMS platforms (e.g., Google Classroom, Moodle) for classroom-wide emotion tracking	2 Months	Enables educators to monitor class-wide emotional trends and intervene early
Phase 3	Personalized long-term learning plans based on emotional history and progress patterns	4 Months	Improves learning outcomes and re-engagement for struggling students
Phase 4	Advanced analytics dashboard, multilingual support, and mobile application	6 Months	Expands accessibility, enhances user experience, and supports global learners